Area between two curves

Let $f(x)$ and $g(x)$ be two continuous functions defined on x in the interval $[a, b]$. Also $f(x) > g(x), a \le b$

Then the area between these two curves from $x = a$ to $x = b$, is

$$A = \int_a^b [f(x) - g(x)]dx.$$

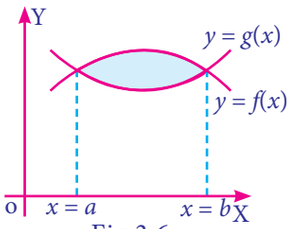


Fig.3.6

Example 3.1

Find the area bounded by $y = 4x + 3$ with x- axis between the lines $x = 1$ and $x = 4$

Solution:

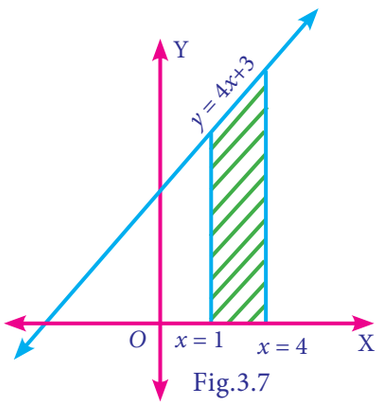


Fig.3.7

Area $= \int_1^4 y dx$

$= \int_1^4 (4x + 3)dx$

$= \left[2x^2 + 3x\right]_1^4 = 32 + 12 - 2 - 3$

= 39 sq.units

Example 3.2

Find the area of the region bounded by the line $x - 2y - 12 = 0$, the y-axis and the lines $y = 2$, $y = 5$.

Solution:

$x - 2y - 12 = 0$

$x = 2y + 12$

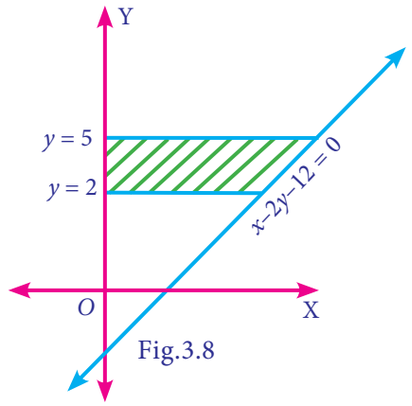


Fig.3.8

Required Area

$= \int_2^5 x\,dy$

$= \int_2^5 (2y + 12)dy = \left[y^2 + 12y\right]_2^5$

$= (25 + 60) - (4 + 24) = 57$ sq.units



The area bounded by the line $y = mx+c$ with x-axis between lines $x = 0$ and $x = a$ is

$= \left|\frac{1}{2}\right.$ [a][(value of y at $x = 0$) + [(value of y at $x = a$)]$\left.\right|$

Example 3.3

Find the area of the region bounded by the parabola $y = 4 - x^2$, $x - axis$ and the lines $x = 0, x = 2$.

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